一. 填空. 2022001300077 智能. 薛润泽.

1. $\frac{2}{3}e^{-2}$ 2. $\frac{19}{27}$

3. $f_Y(y)=\begin{cases}\frac{1}{4\sqrt{y}}, & 0<y<4\\ 0, & \text{其它}.\end{cases}$

二. 选择.

1. A. 2. D. 3. A.

4. D.

1. (1) $\int_{-\infty}^{\infty} P(x)dx = 1$

得 $\int_{-1}^{1} \frac{A}{\sqrt{1-x^2}} dx = A \cdot \arcsin x \Big|_{-1}^{1} = 1$

$A \cdot (\frac{\pi}{2} + \frac{\pi}{2}) = 1 \quad A = \frac{1}{\pi}$

(2) $P(-\frac{1}{2} < X < \frac{1}{2}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\frac{1}{\pi}}{\sqrt{1-x^2}} dx = \frac{1}{3}$

(3) $F(X) = \int_{-\infty}^{x} P(x) = \begin{cases} 1 & , x \geq 1 \\ \frac{1}{\pi} \arcsin x + \frac{1}{2} & , -1 < x < 1 \\ 0 & x \leq -1 \end{cases}$

2. $P(X \leq 50) = P(\frac{X-\mu}{4} \leq \frac{50-40}{4}) = P(\frac{X-\mu}{4} \leq 2.5)$

$= \Phi(2.5) = 0.99379$ ~~≈ 0.99~~

(2) 连续 10个月不超过 50mm.

$P = (0.99379)^{10} \approx 0.9396$

3. $P(X=8) = \frac{\lambda^8}{8!} \cdot e^{-\lambda}$

$P(X=10) = \frac{\lambda^{10}}{10!} \cdot e^{-\lambda}$

$\frac{P(X=8)}{P(X=10)} = \frac{90}{\lambda^2} = 2.5$.

$\lambda = 6$

(1) ~~$P(X=8) = P(X \geq 8) - P(X \geq 9) =$~~

$P(X=8) = \frac{6^8}{8!} \cdot e^{-6} \approx 0.1033$

$P(X=10) = \frac{6^{10} \cdot e^{-6}}{10!} \approx 0.0413$

(2) ~~$P(X \geq 1) = \sum_{k=1}^{\infty} \frac{6^k}{k!} e^{-6}$~~

$P(X=0) = \frac{6^0}{0!} \cdot e^{-6} = e^{-6} \approx 0.002479$

$P(X \geq 1) = 1 - P(X=0) = 1 - e^{-6} \approx 0.997521$

(3) $P(X=1) = 6 \cdot e^{-6} = 0.014872$

$P(X=2) = \frac{6^2 \cdot e^{-6}}{2!} = 0.044617$

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= 0.061968$

≈ 0.062

4. $F_Y(y) = P(Y \le y) = P(e^X \le y)$

$$= \begin{cases} P(X \le \ln y) & y \ge 1 \\ 0 & y < 1 \end{cases}$$

当 $y \ge 1$ 时 $F_Y(y) = P(X \le \ln y) = \int_0^{\ln y} e^{-x} dx$.

$f_Y(y) = F_Y'(y) = \frac{1}{y} \cdot e^{-\ln y} = \frac{1}{y^2}$

所以 $f_Y(y) = \begin{cases} \frac{1}{y^2}, & y \ge 1 \\ 0, & y < 1 \end{cases}$